The unit sphere is covered by the rotations of the representative set

The Open Logic Project contributors

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Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel vitalicover of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the equivalence relation on the sphere identifying points reachable by a rotation through an angle in the countable group
- the choice-of-representatives set C with one element per equivalence class
- rotations as functions on the sphere

Lemma 0.1. $S^1 = \bigcup_{\rho \in \mathrm{SO}(2)} \mathrm{im}(\rho) C$.

Proof. Fix $s \in S^1$; there is some $r \in C$ such that $r \in [s]{\sim}$, i.e., $r \sim s$, i.e., $\rho(r) = s$ for some $\rho \in \mathrm{SO}(2)$. □

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